Classification

Qualitive data; categories, classes

So far we have one-hot encoded categorical data and split our models.

Also logistic regression using Maximum Likelihood Estimation.

logistic : [0, 1]

$$C(\beta) = \sum_{i=1}^{n} \left(-y_i \ln p(x_i)\right) - (1-y_i) \ln (1-p(x_i))$$

$$\Rightarrow \quad P(Y_i = 1 \mid X_i) = \hat{p}(x_i) = \frac{1}{1+e^{-x\beta}}$$

$[0, 1]$

<u>multinomiell</u> $\quad \dfrac{P(y_i = k \mid X_i = x)}{P(y_i = K \mid x_i = x)}$

Multiclass:

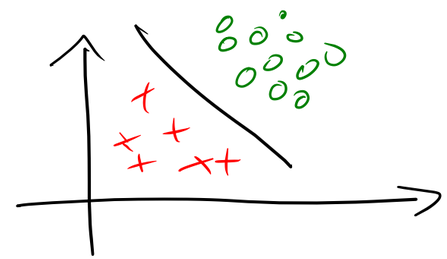
One versus All (one vs rest) OvA

Train several classifiers (one per kategory)
and choose the one with highest p-value.

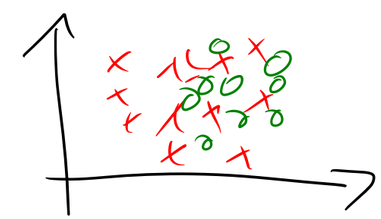
softmax $\left(\frac{Pr(Y=k \mid X=x)}{Pr(Y=k \mid X=x)} \right)$

One versus One OvO

Train a classifier for each kombination of
outcomes; ie [class A vs class B, class A vs C, B vs C, ...]



Logistic regression
less good with
large separations



Logistic regression good!

Linear Discriminants, Quadratic ...

LDA , QDA

Naive Bayes classifier.

Makes very strong assumptions:

Independence between all predictors.

Normal distribution on <u>everything</u>.

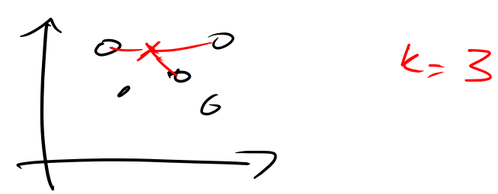
$$P(Y=k \mid X=x)$$

$\rightarrow$ Naive Bayes Classifier

Can also use Bernoulli, Neg-gaussian and some more distribution.

K-NN

k-nearest neighbours



"softmax" over k nearest samples.

Also used in image analysis, specifically recognizing handwritten digits.

Similarity

ML systems need numerical data.

So to be able to measure similarity in

$$Y_i = g^{-1}(X\beta)$$

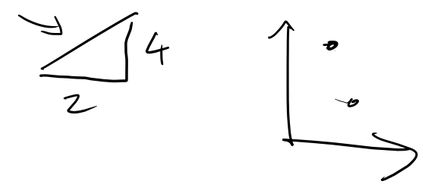
Similar y's must have similar $X_i\beta_i$.

If so, then the following measures distance

- $|a+b| = \ell_1$-norm or taxi-cab distance

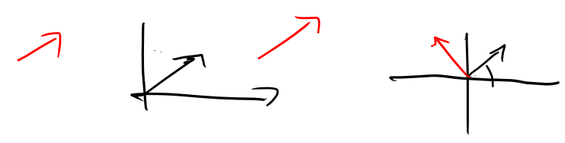
eg: $(x,y) \to |x+y|$ (manhattan distance)

$\sqrt{a^2+b^2}$ = l_2-norm, euclidean distance



rather than a normal cartesian system (normal coordinates) we can use polar coordinates.

Note: vectors are position-independent



Only the angle is now relevant!

$$\frac{u \cdot v}{|u|\,|v|} = \cos\alpha$$

<u>cosine similarity</u>

Text is favourably compared using cosine-similarity,
but we need <u>vectorization</u>

Classically: TF-IDF
Modern: token embeddings in LLMs

Term Frequency / Inverse Document Frequency

Most of these techniques are instance-based (ie don't generalise well)

eg. If you've trained on a corpus of technical manuals, you won't be able to classify fantasy books.